Table 2.6 Different types of coordinate reference systems that are important when building digital representations of a physical system.

System	Description	Key use	Integration challenges
Local coordinate systems	Site-specific frameworks using an origin point and axes aligned to project features (for example building grids)	Engineering drawings and CAD models; intuitive for site design and calculations (for example a facility layout)	Coordinates are only meaningful locally, complicating integration with external data
Geographic coordinate systems (GCS)	Spherical systems representing points on Earth's surface (spheroid) using Latitude and Longitude	GPS data (often using the WGS84 datum / EPSG:4326), satellite imagery, and global asset tracking	Angular nature complicates engineering calculations (distances, areas) due to Earth's curvature
Geocentric coordinate systems	3D Cartesian coordinates with the origin (0,0,0) at the center of the Earth	Earth-Centred, Earth-Fixed (ECEF) / EPSG:4978 is common. Used for direct calculation of 3D distances and vectors between any two global points	N/A
Projected coordinate systems (PCS)	Techniques that flatten the Earth's spheroid onto a 2D plane	Web Mercator (online mapping standard, Google Maps) and Universal Transverse Mercator (UTM)	PCS inherently distort shape or area; for example, Web Mercator exaggerates size away from the Equator

Figure 2.14 shows the same point on the earth's surface represented in a geographic coordinate system (latitude, longitude) on the left, and in a projected coordinate system (EPSG:54050). Notice that the projected coordinate system coordinates are presented as meters from an origin point on a 2D plane, making calculations simpler than those required with with the angular nature of the spherical geographic coordinate system.